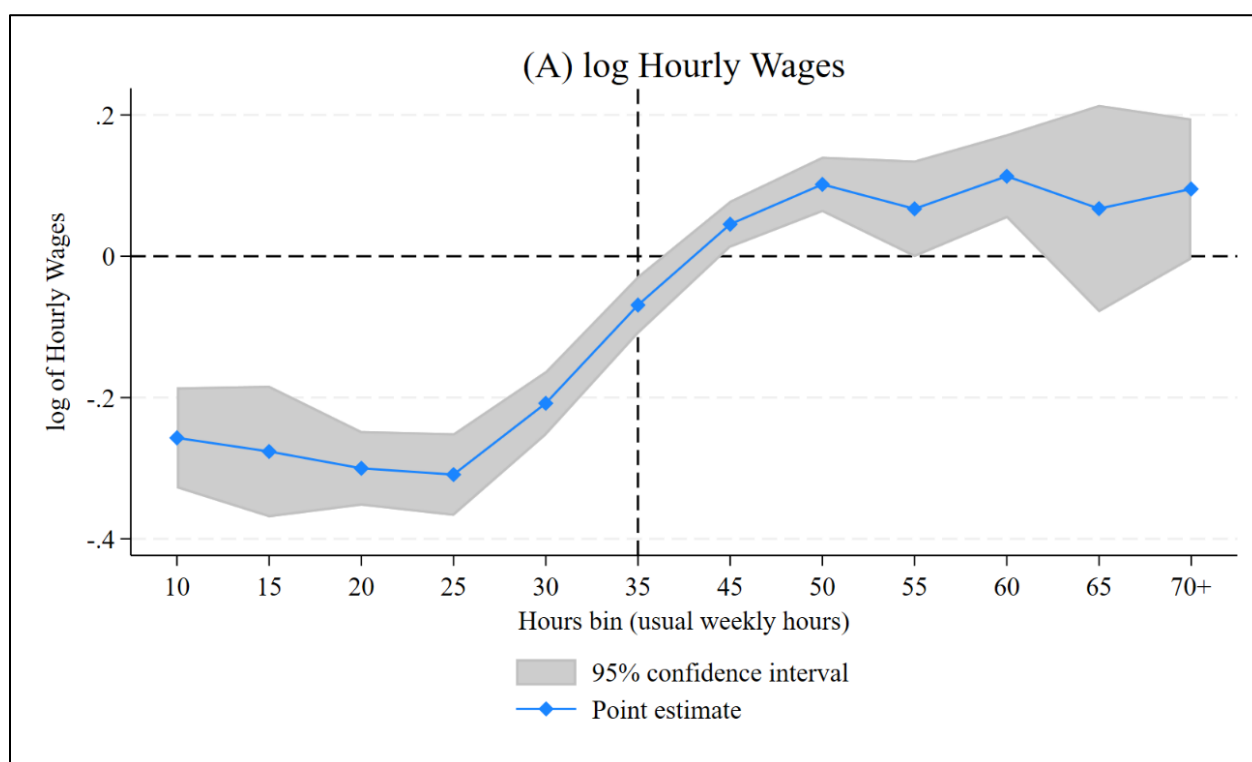


Research Assistant Position: STATA Test

Submitted by: Lavanya Goswami
29 January 2026

1. Replication figure III (Panel A)



Cross-Sectional Relationship between Wages and Hours for Men, aged 25-64.

Source: CPS ORG 2019; N = 10,304.

2. Sample Restrictions Table

Restriction	Code for restriction	Number of observations	Observations dropped	(New sample)% of original
Starting sample	Starting sample – Year = 2019, IPUMS (ORG)	14,24,063	0	100.00%
Working age restriction	keep if age>=25 & age<=64	7,36,310	6,87,753	51.70%
Only Males	keep if sex==1	3,55,707	3,80,603	24.98%
Drop imputed weekly earnings	drop if qearnwee==4	3,31,464	24,243	23.28%
Drop imputed weekly working hours	keep if quhrsworkorg==0	3,31,456	8	23.28%
Drop imputed hourly wage	keep if qhourwag==0	3,29,839	1,617	23.16%
Outgoing Rotation Group sample	keep if mish==4 mish==8	64,271	2,65,568	4.51%
Missing working hours	drop if uhrsworkorg>=998	13,344	50,927	0.94%
Working hours less than 10 (main job)	drop if uhrsworkorg<10	13,241	103	0.93%
Not enrolled in school or college	keep if schllcoll==5	10,337	2,904	0.73%
Not self-employed, unpaid or missing	drop if classwkr<15 classwkr>28	10,337	0	0.73%
Less than half of avg federal min wage	drop if hourwage <4	10,304	33	0.72%
Missing hourly wage	drop if hourwage==999.99	10,304	0	0.72%
Missing weekly earnings	drop if earnweek==9999.99	10,304	0	0.72%
Missing union membership details	drop if union==0	10,304	0	0.72%
Missing industry type	drop if ind1950 == 0 ind1950 == 997 ind1950 == 998	10,304	0	0.72%
Missing marital status details	drop if marst==9	10,304	0	0.72%
	FINAL SAMPLE	10,304	0	0.72%

3. Variable Definitions Table

IPUMS Variables with short descriptions (for full descriptions see appendix)

Concept	IPUMS Variable	Description
Year	YEAR	YEAR reports the year in which the survey was conducted. YEARP is repeated on person records.
Month	MONTH	MONTH indicates the calendar month of the CPS interview.
ORG	MISH	MISH indicates the number of times (from 1 to 8) occupants of a housing unit have been interviewed for the CPS.
State code	STATEFIP	STATEFIP identifies the household's state of residence, using the Federal Information Processing Standards (FIPS) coding scheme, which orders the states alphabetically.
Metropolitan	METRO	METRO indicates whether a household resided in a metropolitan area.
Age	AGE	Age gives each person's age at last birthday.
Sex	SEX	SEX gives each person's sex.
Race	RACE	Racial categories in the CPS have been more consistent than racial categories in the census.
Marital Status	MARST	MARST gives each person's current marital status, including whether the spouse was currently living in the same household.
Industry Code	IND1950	IND1950 recodes information contained in the variable IND into the 1950 Census Bureau industrial classification system.
Class of worker	CLASSWKR	CLASSWKR indicates whether a respondent was self-employed, was an employee in private industry or the public sector, was in the armed forces, or worked without pay in a family business or farm.
Educational attainment	EDUC	EDUC indicates respondents' educational attainment, as measured by the highest year of school or degree completed.
Current enrollment	SCHLCOLL	SCHLCOLL indicates whether respondents age 16 to 24 (or 16 to 54 for ASEC 2013 onward) were enrolled in high school or college during the previous week, and, if so, whether they were enrolled full- or part-time.
Weight on Earnings	EARNWT	EARNWT is a person-level weight that should be used in any analysis including one of the following variables: EARNWEEK, HOURWAGE, PAIDHOUR, UNION, UHRSWORKORG, WKSWORKORG, ELIGORG, and OTPAY.
Quality of Earning Variables	QEARNWEE	Data quality flag for EARNWEEK.
Hourly Wage	HOURWAGE	HOURWAGE reports how much the respondent earned per hour in the current job, for those workers paid an hourly wage (and coded as "2" in PAIDHOUR).
Paid hourly - dummy	PAIDHOUR	PAIDHOUR is a dichotomous variable indicating whether the respondent was paid by the hour for their current job or not.
Union membership	UNION	UNION indicates whether, for the current job, the respondent was: 1) a member of a labor union or employee association similar to a union; 2) not a union member but covered by a union or employee association contract; or 3) neither a union member nor covered by a union contract. Researchers should use the EARNWT weight with this variable.
Usual weekly earning	EARNWEEK	EARNWEEK reports how much the respondent usually earned per week at their current job, before deductions.
Usual weekly working hours	UHRSWORKORG	UHRSWORKORG reports the total number of hours the respondent who is paid hourly usually works per week at their main job.
Quality of working hours variable	QUHRSWORKORG	Data quality flag for UHRSWORKORG.
Quality of hourly wage variable	QHOURWAG	Data quality flag for HOURWAGE.

4. Write-up

(a) Methodology and Approach

The objective of this task is to understand the relationship between hourly wages and usual weekly hours worked. We want to check whether the inverted u-shape graph obtained in previous works also exists in more recent data. The dataset used is IPUMS outgoing rotational group from the year 2019. We restrict the sample as described in the paper. The method used is an OLS regression model, with suitable socio-economic controls. The dependent variable is log of hourly wages while the main independent variable of interest is a categorical variable of weekly hours worked. The weekly hours worked are binned together in 5-hour slots and top coded at 70. Thus, the coefficients measure the percentage increase in hourly wages associated with a 5-hour increase in weekly working hours, ceteris paribus. The values predicted by this regression are hence plotted on the graph.

(b) Key differences from original paper

The original paper used a larger sample size, over multiple years, making the analysis robust to macroeconomic-shocks, and has more consistent estimates.

In the replicated graph, weekly hours worked increase with the same propensity as observed in the original graphs, and peak similarly at the 50-hour bin. However, wages seem to flatten out at higher weekly hours worked instead of displaying a clear inverted u-shape graph.

While there is a dip at the 65-hour bin, it is neither statistically significant nor as great in magnitude as expected. The relation between log hourly wages and usual weekly hours is still non-monotonic. The replication neither confirms nor denies the existence of a wage penalty for working beyond 50 hours, which was a key result of the original paper.

(c) Challenges encountered and solutions

- The variable SUBMINWAGE, an indicator for whether the wage of an individual is less than the federal minimum wage was missing for 2019. Instead, I used the average federal minimum wage cited [in this document](#).
- Following the 5-hour bin coding as defined in original paper, the confidence interval in the 70-hour bin exploded due to very low samples. Thus, the top coding was shifted from 85+ to 70+.

(d) AI Declaration

I used ChatGPT to debug syntax errors (for example: brackets in for loops), write code for variable labels (in case of multiple and tricky categories like industry codes) and to write Python code to scrape descriptions of IPUMS variables in a excel file. I also considered suggestions for data cleaning restrictions.

5. Critical Thinking Questions

Question 1: Women Extension (100-150 words) What methodological considerations would arise if extending this to women? Address selection and methodological issues.

Women's labour force participation rate tends to be lower and even within the restricted sample, working hours tend to be lower than men (for instance, women may engage in more part-time work). We would thus have a much smaller sample size and the distribution would be right-skewed. In fact, women's labour force participation itself tends to be hard to define since household production for self is not counted as labour force participation despite having economic value.

Women's elasticity of labour with respect to wages varies, depending on the economic and cultural context. Care-giving responsibilities at home may result in an attenuated response to higher wages. Women's work may be conditional on macroeconomic shocks and household-level shocks, thus confounding the productivity relationship.

Question 2: Cross-Country Extension (100-150 words) What challenges would arise in comparing US and South Korea? Consider data comparability, institutional differences, and variable harmonization.

Comparing this data across US and South Korea would require all income variables to be adjusted for purchasing power parity. Similarly, harmonized variables would be required for control variables like educational qualifications and type of industry. The effects of unionisation would be different in these countries based on labour laws.

Differences in sample selection methods, question definitions and methods of survey may also complicate the comparability of the data.

Appendix

IPUMS Variables with full descriptions (scrapped using Python from official website)

Concept	IPUMS Variable	Description
Year	YEAR	YEAR reports the year in which the survey was conducted. YEARP is repeated on person records.
Month	MONTH	MONTH indicates the calendar month of the CPS interview.
ORG	MISH	MISH indicates the number of times (from 1 to 8) occupants of a housing unit have been interviewed for the CPS. Household members are interviewed for four consecutive months, excluded for eight months, and then included for four more consecutive months. On first interview, a household has a value of 1 for MISH. Households returning to the sample after an 8-month hiatus have a value of 5, and those which have completed their last interview have a value of 8. Persons with codes of 4 or 8 in MISH are said to be in "outgoing rotation groups," because they will not be interviewed during the following month. For 1962-1967, the same information is given in ROTATE, a person-level variable. IPUMS CPS has identified an error in the month-in-sample values in the original data in the 2016, 2018, 2020, and 2022 CPS-ASEC files. Records in the Census Bureau CPS-ASEC PUMS files in these years have month-in-sample values that do not correspond to these record's values in the March basic file. IPUMS CPS corrects the March basic monthly respondent's month-in-sample values in the 2016, 2018, 2020, and 2022 CPS-ASEC to match the March basic monthly file values. Month-in-sample values for ASEC oversample records are not adjusted. The fix is as follows: Original month-in-sample of 1 become MISH of 5Original month-in-sample of 2 become MISH of 6Original month-in-sample of 3 become MISH of 7Original month-in-sample of 4 become MISH of 8Original month-in-sample of 5 become MISH of 1Original month-in-sample of 6 become MISH of 2Original month-in-sample of 7 become MISH of 3Original month-in-sample of 8 become MISH of 4
State code	STATEFIP	STATEFIP identifies the household's state of residence, using the Federal Information Processing Standards (FIPS) coding scheme, which orders the states alphabetically. In 1973-1975 ASEC samples, all households in the Anaheim-Santa Ana-Garden Grove, CAMEAREAS are coded as Michigan-Wisconsin for STATEFIP in the original data. As there is insufficient geographic information in the public use data to determine which variable is in error, this mistake has been left un-recoded.
Metropolitan	METRO	METRO indicates whether a household resided in a metropolitan area. For households within metropolitan areas, METRO also indicates whether the housing unit was inside or outside of a central/principal city. The Census Bureau adds this information itself and does not collect it directly from respondents.
Age	AGE	Age gives each person's age at last birthday.
Sex	SEX	SEX gives each person's sex.
Race	RACE	Racial categories in the CPS have been more consistent than racial categories in the census. Up through 2002, the number of race categories ranged from 3 (white, negro, and other) to 5 (white, black, American Indian/Eskimo/Aleut, Asian or Pacific Islander, and other). Beginning in 2003, respondents could report more than one race, and the number of codes rose to 21, and then up to 26 codes in 2013.
Marital Status	MARST	MARST gives each person's current marital status, including whether the spouse was currently living in the same household.
Industry Code	IND1950	IND1950 recodes information contained in the variable IND into the 1950 Census Bureau industrial classification system. Developed to enhance the comparability of industry data in historical U.S. census samples in IPUMS-USA, IND1950 also provides a consistent set of industry codes for IPUMS-CPS from 1968 forward. For general discussion of the IND1950 variable, users should consult the IPUMS-USA documentation. For discussion of the CPS data on industry that are recoded into IND1950, see IND.
Class of worker	CLASSWKR	CLASSWKR indicates whether a respondent was self-employed, was an employee in private industry or the public sector, was in the armed forces, or worked without pay in a family business or farm. Workers with multiple sources of employment were classified according to the job in which they worked the most hours. For persons employed at the time of the survey, CLASSWKR relates to the respondent's job during the previous week. Respondents who were not employed during the previous week reported the most recent job. The CLASSWLY variable deals with the longest job held during the previous calendar year. In the original CPS public use data, members of the armed forces are coded as N.I.U. (Not in Universe), because they are not part of the civilian labor force.
Educational attainment	EDUC	EDUC indicates respondents' educational attainment, as measured by the highest year of school or degree completed. Note that completion differs from the highest year of school attendance; for example, respondents who attended 10th grade but did not finish were classified in EDUC as having completed 9th grade. EDUC is a combination of two other variables, HIGRADE and EDUC99, which measure educational attainment in different ways. HIGRADE is available for years prior to 1992 and gives the respondent's highest grade of school or year of college completed. EDUC99 is available beginning in 1992 and classifies high school graduates according to their

		highest degree or diploma attained. General and detailed codes are not yet available for IPUMS-CPS, but one can construct the general version of EDUC by reading only the first two columns of EDUC.
Current enrollment	SCHLCOLL	SCHLCOLL indicates whether respondents age 16 to 24 (or 16 to 54 for ASEC 2013 onward) were enrolled in high school or college during the previous week, and, if so, whether they were enrolled full- or part-time. College or high school students who were currently on holiday or seasonal vacation were to answer yes, but those not taking classes during summer vacation were to answer no. Interviewers first asked whether the person was enrolled in school during the previous week, then determined whether the person was attending high school or attending a college or university, and finally asked whether the person was a full-time or part-time student. In IPUMS-CPS, these responses are combined into the single variable SCHLCOLL.
Weight on Earnings	EARNWT	EARNWT is a person-level weight that should be used in any analysis including one of the following variables: EARNWEEK, HOURWAGE, PAIDHOUR, UNION, UHRSWORKORG, WKSWORKORG, ELIGORG, and OTPAY. For any other analysis using ASEC data, researchers should use ASECWT or for analyses of non-ASEC data, WTFINL. Individuals in the 6 rotation groups that were not asked the "earner study" questions (covering EARNWEEK, HOURWAGE, PAIDHOUR, UNION, UHRSWORKORG, WKSWORKORG, ELIGORG, and OTPAY) have a value of zero for EARNWT. Even in the 2 rotation groups where "earner study" questions were fielded, members of the armed forces have a value of zero for EARNWT. According to Technical Paper 66, issued jointly by the Census Bureau and the Bureau of Labor Statistics, individuals eligible for the earner study are civilians age 15 and older in rotation groups 4 or 8 who are not self-employed. In the data itself, this is true in all samples up to and including those in 1997. From 1998 to 2010, only individuals age 16 and older in rotation groups 4 or 8, and are not self-employed, are in universe for this variable. This universe also applies to basic monthly samples after 2010. In ASEC samples after 2010, the data reflect the published universe. In any given month, approximately 1/4 of the CPS sample is in the earner study and each household should appear in the earner study exactly twice. Based on documentation from Unicon and NBER, and after an inspection of the original CPS data, we recommend that users impose the CPS eligibility restrictions in any analyses of earner study variables.
Quality of Earning Variables	QEARNWEE	Data quality flag for EARNWEEK.
Hourly Wage	HOURWAGE	HOURWAGE reports how much the respondent earned per hour in the current job, for those workers paid an hourly wage (and coded as "2" in PAIDHOUR). HOURWAGE is one of the Outgoing Rotation/Earner Study questions. Researchers should use the EARNWT weight with this variable. Users should note that HOURWAGE originally had two implied decimal places, but was revised so that the command files provided by IPUMS divide HOURWAGE by 100. Amounts are expressed as they were reported to the interviewer; users must adjust for inflation using Consumer Price Index adjustment factors or use the Adjust Monetary Values tool. Inflation-adjusted versions created with the Adjust Monetary Values tool will use the survey year (i.e., YEAR) as the reference year.
Paid hourly - dummy	PAIDHOUR	PAIDHOUR is a dichotomous variable indicating whether the respondent was paid by the hour for their current job or not. Researchers should use the EARNWT weight with this variable. PAIDHOUR is one of the Outgoing Rotation/Earner Study questions.
Union membership	UNION	UNION indicates whether, for the current job, the respondent was: 1) a member of a labor union or employee association similar to a union; 2) not a union member but covered by a union or employee association contract; or 3) neither a union member nor covered by a union contract. Researchers should use the EARNWT weight with this variable.
Usual weekly earning	EARNWEEK	EARNWEEK reports how much the respondent usually earned per week at their current job, before deductions. Interviewers asked directly about total weekly earnings and also collected information about the usual number of hours worked per week and the hourly rate of pay at the current job. The figure given in EARNWEEK is the higher of the values derived from these two sources: 1) the respondent's answer to the question, "How much do you usually earn per week at this job before deductions?" or 2) for workers paid by the hour (and coded as "2" in PAIDHOUR), the reported number of hours the respondent usually worked at the job, multiplied by the hourly wage rate given in HOURWAGE. Amounts are expressed as they were reported to the interviewer; users must adjust for inflation using Consumer Price Index adjustment factors or use the Adjust Monetary Values tool. Inflation-adjusted versions created with the Adjust Monetary Values tool will use the survey year (i.e., YEAR) as the reference year. EARNWEEK is one of the Outgoing Rotation/Earner Study questions. Researchers should use the EARNWT weight with this variable.
Usual weekly working hours	UHRSWORKORG	UHRSWORKORG reports the total number of hours the respondent who is paid hourly usually works per week at their main job. UHRSWORKORG is one of the earner study/outgoing rotation group questions. Private wage or salaried workers are asked the periodicity for which they are paid. Workers who are paid by the hour, excluding those who are self-employed, are asked the hourly wage, and then are asked how many hours they usually work per week at that rate. Researchers should use the EARNWT weight with this variable. Researchers should note that PAIDHOUR, which identifies workers paid hourly, is more inclusive than the universe for UHRSWORKORG. This stems from the fact that PAIDHOUR is the combination of two variables that assess hourly work status. The first establishes periodicity of pay, and the second asks if an hourly rate can be given instead if respondents did not report hourly pay. Only those who reported an hourly pay status in the first question are in universe for UHRSWORKORG. UHRSWORKORG differs from UHRSWORKLY because the period of reference for UHRSWORKLY is the preceding year, whereas UHRSWORKORG refers to the current usual number of hours worked per week. Additionally, UHRSWORKORG differs from UHRSWORK1 in that UHRSWORK1 does not have the qualifier "at this rate" in the question. Note that for basic samples 1982-1988 and ASECs 1982-1987, the outgoing rotation usual hours question wording matches that of UHRSWORK1. Thus, this variable is included in UHRSWORK1 for those years, not in UHRSWORKORG. UHRSWORK1 specifically gives the number of hours worked last week at the main job, regardless of whether this was the usual number of hours worked. UHRSWORKORG is one of the Outgoing Rotation/Earner Study questions. See the Hours Worked Variables Notes for an overview of the different actual and usual hours worked variables available.
Quality of working hours variable	QUHRSWORKORG	Data quality flag for UHRSWORKORG.
Quality of hourly wage variable	QHOURWAG	Data quality flag for HOURWAGE.